MLPC 2026 Task 3: Data Exploration

Monday, March 23, 2026

Context

Kepler Intelligent Audio Labs (KIAL), a soon-to-be-founded innovative AI startup, aims to collect an audio recording dataset from domestic environments with strong temporal annotations for a set of pre-defined target sound event classes. The goal is to capture a diverse range of sound events across varied domestic settings in order to reflect real-world acoustic conditions and enable robust sound event detection systems for different customer use cases (see Figure 1 for a schematic illustration of sound event detection).

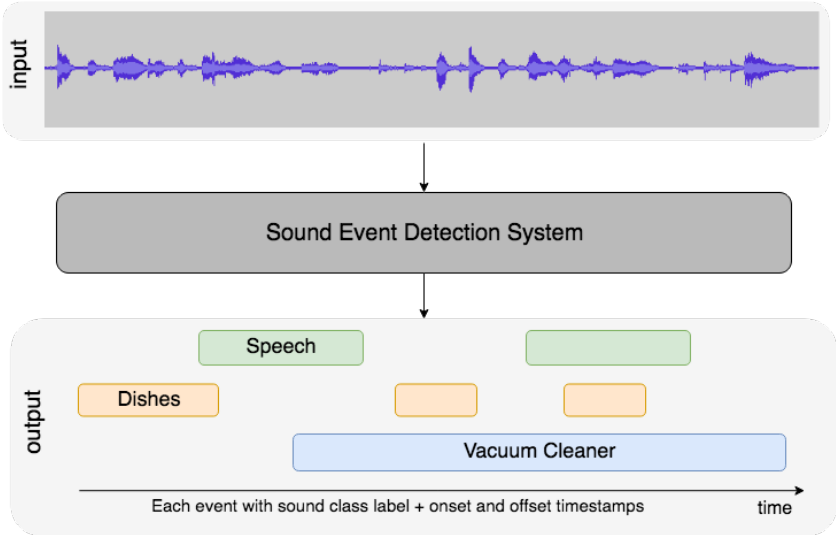

Figure 1: **Sound Event Detection (SED)** systems take an audio recording as input and predict acoustic events of interest (here: Speech, Dishes, and Vacuum Cleaner), *including* their temporal onsets and offsets. Figure taken from the DCASE website.

To realize their vision, KIAL has identified the following milestones:

1. **Data Collection:** Define the target sound event classes and collect audio recordings containing these events in representative domestic environments. (✓)
2. **Data Annotation:** Human annotators label the recordings by identifying sound events and marking their temporal onsets and offsets. (✓)
3. **Data Analysis:** Validate and explore the dataset to understand its structure, variability, and potential challenges. (← we are here)
4. **Model Training & Selection:** Train and optimize machine learning models to map audio features to sound event labels and select the best-performing approach.
5. **Customer Validation:** Evaluate the system on a secret test set to obtain an unbiased estimate of detection performance.

Task Outline: Data Exploration (25 points)

Deadline: Thursday, April 16th, 23:59
Submission: Submit your team's report and slides via Moodle.
Group Work: Both team members are required to contribute to every phase of the project. Talk to your team partner early and decide on a distribution of workload, a schedule, and deadlines. *Check in often.* Plan enough time to help your colleague if they struggle to complete a task. Let us know early if your teammate is unresponsive.

This task is mandatory. Content and formal requirements are outlined below.

1 Report (max. 22 points)

For the first part of this assignment, you are required to write a report based on the template provided on Moodle. Your report must address *all of the following points*:

1. **Verify Annotations & Qualitative Analysis** (8 points): Your team is required to compare and verify annotations for 10 audio recordings using Label Studio (5 reviews per person)[1].

   (a) **Verify Annotations:** Your team has been assigned audio recordings with at least two annotations to be reviewed via Label Studio. Verify the correctness of the annotations for each audio recording. **Important: The verification can only be done once!** *Carefully read the verification guide provided in Appendix A before you start the review.* Take notes and screenshots for questions 1.b. and c.[2]
   Expected Outcome: You have thoroughly verified annotations of all assigned audio recordings via Label Studio, taken notes and screenshots to answer the following questions.

   (b) **Summary of Disagreements:** What kinds of disagreements between annotators (i.e., valid differences of selected temporal boundaries or class annotations) and clear violations of the annotation task description did you observe? Does disagreement always imply that at least one annotator violated the annotation rules?
   Expected Answer: Give a concise summary of disagreement and violation trends you've observed.

   (c) **Case Study:** Present a single case in your report that illustrates a clear violation of the annotation rules or a case where two annotators produced significantly disagreeing annotations (valid or invalid).
   Expected Answer: Provide the filename (metadata section in Label Studio), a (side-by-side) visualization of the annotation(s) (e.g., a spectorgram plot), a textual description of the violation/ disagreement and an interpretation.

2. **Annotations** (6 points): Use the annotations in the feature files (npz-files) to address the following points *quantitatively.* A detailed description of the feature files, which also contain the feature-aligned annotations, is provided in Appendix B. *Read the dataset description.*

   (a) **Annotator Agreement:** Quantify the overall annotator agreement. Is disagreement higher for certain classes than for others? Do the temporal annotations agree with the target class tags specified by the uploading user?
   Expected Answer: Explain how you quantified the agreement. Present an estimate of the general agreement and the class-wise agreement. Give a textual interpretation of your results. Include a visualization or a table if appropriate.

   (b) **Convert Annotations to Labels:** Derive class labels for every time step in the annotation array (i.e., aggregate the annotation across all annotators, and derive binary prediction targets). What assumptions does your method of aggregation and binarization rely on? Under which conditions could the resulting labels be incorrect?
   Expected Answer: Describe your method of aggregation and binarization and explain your reasoning and potential imperfections.
   Hint: In addition to the temporal annotations, the feature file also provides additional metadata

[1] Review tasks will assigned once the Annotation period is over, i.e., on March 26
[2] In case forgot to take notes during verification, use the raw audio files provided on our Moodle and `annotations.csv`.

like the target class tags specified during data collection and a boolean mask specifying whether the annotator is the same as the collector of the recording.

(c) **Label Characteristics:** Describe characteristics of the resulting labels (e.g., frequency of class events, class-wise average or total duration of events, co-occurrence patterns in the same recording or with recording locations, ...).
Expected Answer: Describe which characteristics you looked into and how you quantified them. Provide the results and a textual interpretations. Include visualizations if appropriate.

3. **Audio Features** (6 points + 2 bonus points): Load and analyze the audio features *quantitatively.* A detailed description of the feature files and their content is provided in the Appendix B. *Read the dataset description.*

   (a) **Metadata:** Check the distribution of recording properties (e.g., placement, recording device types and recording locations). You can either collect the values from the feature files or directly use the `metadata.csv`.
   Expected Answer: Explain which property you looked into. Provide statistics or a visualization and a short interpretation.

   (b) **Feature Statistics:** Calculate basic statistics of the audio feature values (mean, standard deviation, range).
   Expected Answer: Describe which statistics you calculated and provide a compact summary (or a visualization) and an interpretation.

   (c) **Feature Correlation:** Which audio features (or categories of audio features, e.g., Mel-Spectorgram and MFCCs) appear to be correlated?
   Expected Answer: Explain how you analyzed correlation. Quantify the correlation and provide an interpretation or visualization of the results.

   (d) **Bonus Task (2 points): Feature Space Visualization** Select audio features and use a method of down-projection to create a visualization of the feature space. Try to visually identify meaningful clusters by using your class labels or properties given in the metadata (e.g., recording location/ characteristics).
   Expected answer: Explain which features you used and why. Which method of down-projection did you use and why? How did you tune the down-projection's hyperparameters. Visualize the result and highlight selected clusters of classes. Give a short textual interpretation.
   Hints: Use a down-projection (e.g., PCA, UMAP, or t-SNE) to two dimension. Select a high quality subset of the dataset (high annotator agreement and only one class present in the segment).

4. **Conclusions** (2 points): What conclusions can you draw from your analysis for the next phases of the project?

   (a) **Biases:** Which biases were introduced during the data collection and annotation phase?
   Expected Answer: Give at least three examples of biases and describe potential consequences.

   (b) **Conclusion:** Is the dataset suitable for training sound event detectors in the next phase? Which aspects of the dataset will make learning reliable classifier challenging?
   Expected Answer: Give a textual answer. Draw conclusions from your previous findings for the next phases of the project.

   (c) **Broader Impact:** What could be the broader positive and negative societal impact of creating and making such a dataset publicly available? What potential misuse scenarios should be considered? Which measures can creators of such a dataset take to promote ethical use and reduce the risk of misuse?
   Expected Answer: Give a short description of potential impacts and measures to promote ethical use.

The report **must not exceed a page limit of 5 pages with at most 1500 words** (Overleaf: File → Word count). For the bonus task (3.d.), you can additionally use half a page and 150 words. You may add figures into an appendix, however, there's no guarantee that we'll take them into account for grading.

The use of large language models (LLMs) for brainstorming, coding (e.g., implementation of down-projection or creating fancy matplotlib visualizations), or improving the clarity of your writing is explicitly allowed. Critically evaluate all outputs and include only content in your report that you understand and have verified.

All teams are required to include a "Disclosure of LLM and AI Tool Use"-section at the end of their report. This section must briefly *clarify the following points:*

- Whether or not you used any Large Language Models (LLMs) or other AI tools (image generators).
- Which models did you use and for which tasks (e.g., brainstorming solution methods, improving writing, writing code).
- How did you verify the correctness of the LLM-generated content.

This "Disclosure of LLM and AI Tool Use"-section is mandatory for all teams (even if no tools were used) and does not count toward the page limit. Responsible use of LLMs will not affect your grade negatively. However, points will be deducted for severe cases of "AI slop," or failing to include a "Disclosure of LLM and AI Tool Use"-section.

2 Slide Set (3 points)

The complementary slide set should present the results from your written report in a concise manner. More precisely, you will have to answer the questions corresponding to one of the sub-topics outlined in the previous section. The specific topic is determined based on the first letter of your group name, i.e. A for Team Autonomous Architectures, or V for Visionary Agents. To find your topic, determine the according letter, and find your topic in the following list:

First letter of group name	Topic
A-C	Qualitative Analysis
D-G	Annotations
H-Q	Audio Features
R-Z	Conclusion

Table 1: Assignment of topics for the slides based on the first letter our your group name.

The **upper limit for the number of slides you should prepare is 3** (excluding an additional title slide that should contain your group name and the member names).

3 Development Dataset

The development dataset will be available on our Moodle page once the annotation phase is over (March 26th). The format and content of the dataset are described in the Appendix B and in the classroom meeting's recording. Have a look at the slide deck of our first tutorial session (March 30th) to see how the dataset can be loaded.

4 Grading

Each of the four topics given in the task outline above will be evaluated according to the following criteria:

- Thoroughness & Completeness: Have you thought about the problem, and answered every question?
- Clarity: Are the ideas, features, algorithms, and results described clearly? Based on your descriptions, could the reader reconstruct your experiments?
- Presentation: Did you select an appropriate way of communicating your results, e.g., did you use meaningful plots where helpful?
- Correctness: Is the proposed procedure/experiment sound, correct?
- Punctuality: The reports must be submitted in time. No late submissions will be accepted.

You will be awarded all points for the slide set if it addresses the assigned topic, is within the slide limit, and is submitted before the deadline.

5 Summary

- Completing Task 3 is a requirement to pass this course.
- Look at the given questions and answer all of them appropriately in a written report. Make sure to use the report template provided to you via Moodle. Adhere to the given page limit (max. 5 pages with at most 1500 words).
- Add a "Disclosure of LLM and AI Tool Use"-section to your report. This section does not count toward the page limit.
- Create a set of slides tackling the questions of the topic assigned to your group. The topic is determined by the first letter of your group name. Make sure to adhere to the slide limit for this step as well (max. 3 + 1 title slide).
- Upload the written report as well as your slides to Moodle before the deadline.
- You will get a maximum number of 22 points for your written report, and 3 points for the slide set.

A Annotation Verification via Label Studio

Your team has been assigned audio recordings with at least two annotations from different annotators for review in Label Studio. You are free to distribute the workload as you like. Your task is to verify the correctness of the annotations for each audio recording and fix incorrect annotations.

The following guide explains the steps necessary to complete the data verification task. Read the description carefully! *If we find that you did not properly review the annotations, you will be required to repeat this task.*

Step 1: Login to Label Studio

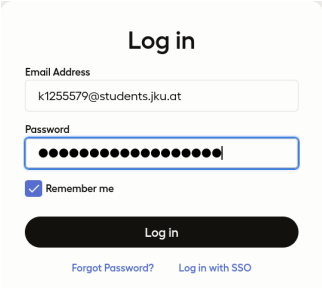

Figure 2: Log into your Human Signal account with your student email.

You should already be familiar with logging into your Human Signal account. Use the link and the credentials below:

- **Link:** https://app.heartex.com/user/login/
- **Username:** `k<matriculation_number>@students.jku.at`, e.g., `k01234567@students.jku.at`
- **Password:** your chosen password.

Important: *Use your generic JKU student email* instead of your personal email to sign up and log in. Everything sent to `k<matriculation_number>@students.jku.at` will be forwarded to the email address set in KUSSS.

Step 2: Select the MLPC2026 Annotation Task

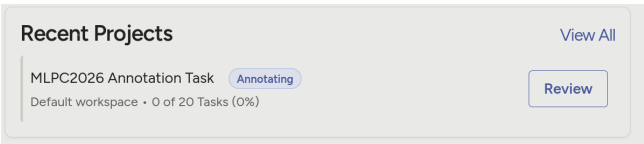

Figure 3: Click on `Review` to open the annotation interface.

The landing page displays a section titled *Recent Projects* which contains one project: MLPC2026 Annotation Task (Fig. 3). Below the title, you can find a short summary of your team's review progress.

Click the `Review` button to open the review interface.

Step 3: Verify Annotations

Your team has been assigned a number of review tasks. A review for a recording counts as completed (for both team members) if all annotations are accepted or fixed and accepted. Rejected annotations will be re-queued

until you've fixed them.

Important: Each review can only be done once! Once you've submitted a review, you cannot update it anymore.

For every assigned review task, complete the following steps:

1. **Get familiar with the recording:** Familiarize yourself with the audio recording and the corresponding metadata and annotations.
2. **Compare the annotations:** Check whether all annotators annotated the recording thoroughly and correctly:
   - Were the classes assigned according to the class definitions (see Appendix of Annotation Task Description)?
   - Were sound events annotated precisely?
   - Did the annotator split regions according to the one-second rule?
   - Did the annotators miss any events?

   The top bar (Fig. 4) of the window shows an overview of all annotations for the given file. Switch in-between the annotations or open a side-by-side comparison by clicking "Compare All" and selecting the "Side-by-side" view (Fig. 5).

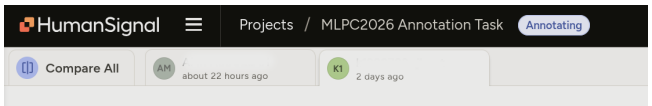

Figure 4: Click the annotation tabs to switch between individual annotations and to modify them. Click "Compare All" → "Side-by-Side" to compare annotations side-by-side.

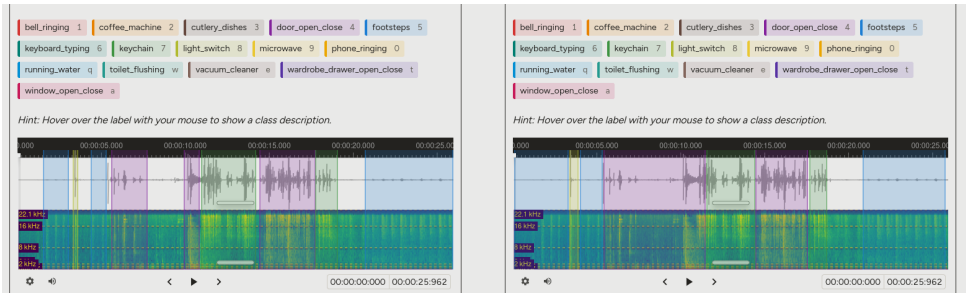

Figure 5: The side-by-side view gives a good overview over annotations by all annotators.

3. **Fix incorrect annotations:** Go back to the individual annotations (select one of the single annotation tabs, Fig. 4) and correct any errors in the annotations by adding, removing, or adjusting regions, or by changing class labels.
   - For example:
     - If both annotators missed a quiet footstep or a brief keychain jingling, add the missing event.
     - If an annotator assigned an incorrect class to a region, e.g. door_opening_closing for a microwave door, change it to the correct class.
     - If an annotator did not split an acoustic event with a pause longer than 1 second into multiple regions, create separate regions.
   - The resulting fixed annotations should roughly agree with each other. However, they do not need to match exactly; slight differences are acceptable (annotators might have perceived onset and offset times differently).
   - If there is strong disagreement in the annotations and you cannot determine which version is correct, fix what you can fix, add a comment and click accept.

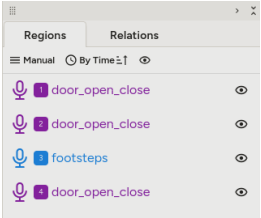

Figure 6: Use the region menu to hide or lock regions.

Hint: Use the region overview in the right side bar to lock or hide regions (Fig. 6).

4. Once you are finished reviewing and have removed all errors:
   - Click "Accept" if the annotation was correct.
   - Click "Fix + Accept" if you made corrections.
   - "Reject" re-queues the recording into your own task queue. Use it only if the recording does not contain audible sounds. If you do, provide a comment.

Once your team has reviewed all assigned audio files, a pop-up will appear (Fig. 7).

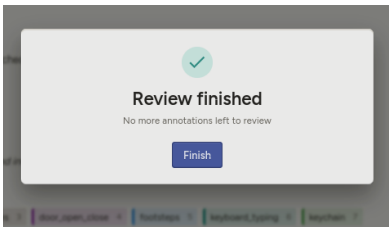

Figure 7: Once you are done, a pop-up window will open.

B The Development Dataset Descriptions

The dataset package has the following structure:

```
MLPC2026_dataset_development.zip
|-- metadata.csv
|-- annotations.csv
\-- audio_features/
    |-- 000001.npz
    |-- 000002.npz
    \-- ...
```

Raw waveforms are not required for this phase of the project but separately available in case you'd like to listen to the recordings.

```
MLPC2026_dataset_development_raw_audio.zip
|-- 000001.wav
|-- 000002.wav
\-- ...
```

B.1 `metadata.csv`

Contains metadata for each audio file. Columns:

Column	Description
`filename`	file identifier
`collector_id`	anonymized identifier for the student who collected the recording
`target_classes`	semicolon separated list of target classes present in the recording (user-provided)
`non_target_classes`	semicolon separated list of target classes present in the recording or `None`
`recording_device`	recording device (user-provided)
`device_placement`	mobile or static
`recording_environment`	semicolon separated list of recording locations
`scene_description`	textual description of the recording scene
`license`	license of the recording

B.2 `annotations.csv`

Contains annotation events in tabular form. Each row corresponds to one annotated event. Files might contain multiple annotation events. Columns:

Column	Description
`filename`	file identifier
`annotator_id`	anonymized identifier; not connectable with collector_id
`annotation`	selected class

`onset`	onset time in seconds
`offset`	offset time in seconds
`is_own_recording`	whether annotator equals collector

B.3 `*.npz` Files

Features and annotations are provided as aggregations over one second windows with a hop size of 0.5 seconds. Filenames are consistent with the filenames specified in metadata.csv and annotations.csv (e.g., `file1.wav` corresponds to `file1.npz`). Load the feature-annotation files with:

```
import numpy as np

features_and_annotations = dict(
    np.load('path/file_name.npz', allow_pickle=True)
)
```

The result is a dictionary holding the features and labels as numpy arrays:

```
{
    # features
    'mfcc_mean': ...,       # [num_segments, feature_dimensionality]
    'mfcc_std': ...,        # [num_segments, feature_dimensionality]
    ...
    # metadata
    'annotations': ...,     # [num_segments, num_classes, num_annotators]
    'start_time': ...,      # [num_segments,]
    'end_time': ...,        # [num_segments,]
    ...
}
```

Below is a detailed description of the feature files' content.

B.3.1 Acoustic Features

Base Key Name	Description	Shape
`zcr`	Zero Crossing Rate: waveform sign changes	[T,1]
`melspect`	Mel spectrogram with 128 mel frequency bands	[T,128]
`mfcc`	Mel-frequency cepstral coefficients (32 coefficients)	[T,32]
`mfcc_d`	First temporal derivative of MFCC (32 coefficients)	[T,32]
`mfcc_d2`	Second temporal derivative of MFCC (32 coeff.)	[T,32]
`flux`	Spectral flux: frame-to-frame spectral change	[T,1]
`flatness`	Spectral flatness: tonality vs. noisiness	[T,1]
`centroid`	Spectral centroid: spectral brightness	[T,1]
`bandwidth`	Spectral bandwidth: spectral spread	[T,1]
`contrast`	Spectral contrast: peak-valley contrast across bands	[T,7]
`rolloff_low`	Low spectral rolloff: low energy cutoff frequency	[T,1]
`rolloff_high`	High spectral rolloff: high energy cutoff frequency	[T,1]

`energy`	Signal energy	[T,1]
`power`	Signal power	[T,1]

Table 4: T corresponds to the length of the segment.

All features are computed for 32ms snippets and then aggregated over one second windows to reduce the temporal resolution. We used the following functions for aggregation; the suffix in the keys' names indicates the aggregation method.

- mean (`*_mean`)
- standard deviation (`*_std`)
- minimum (`*_min`)
- maximum (`*_max`)

To get a feature array use the feature's base name and the suffix, e.g.,

```
features_and_annotations["mfcc_mean"]
```

B.3.2 Temporal Alignment Arrays

`start_time` and `end_time` (in seconds) entries define the segment boundaries for each aggregated feature frame.

Key	Description	Shape
`start_time`	segment start boundaries	[T]
`end_time`	segment end boundaries	[T]

Table 5: T corresponds to the length of the segment.

B.3.3 Annotations

For your convenience, we've converted the annotation in `annotations.csv` into an annotation array that is aligned with the features. Each value represents proportional overlap between an annotation event and a time segment.

Key	Description	Shape
`annotations`	time-aligned annotation indicators for segments and annotators	[T, C, A]

Table 6: T, C, and A correspond to the length of the segment, the number of classes, and the number of annotators.

The values in `annotation` are interpreted as follows:

- `0.0`: no overlap of the segment with an annotation
- `1.0`: full overlap of the segment with an annotation
- values in `(0,1)`: degree of overlap of the segment with an annotation

B.3.4 Additional Label Metadata

Additional metadata which might be useful:

Key	Description
`class_names`	list of classes; corresponds to the class indices in the annotation array; alphabetically sorted, aligned with the second dimension of the annotation array and consistent across all feature files
`annotator_ids`	list of annotator identifiers; aligned with last dimension annotation array
`is_own_recording`	list of boolean flags indicating whether each annotator is the same as the collector of the recording; aligned with last dimension of annotation array
`target_classes`	list of target class labels specified by the uploading user
`non_target_classes`	list of non-target class labels specified by the uploading user
`recording_device`	recoding device; specified by the uploading user
`recording_environments`	list of rooms the recording was captured in; specified by the uploading user
`scene_description`	textual description of the audio's content; specified by the uploading user
`device_placement`	mobile or static; specified by the uploading user

B.4 License

This preliminary version of dataset is restricted for use only within the scope of this course. The features, the metadata and the annotations may not be redistributed, shared, or published outside of this course without explicit permission from the course instructors. Audio recordings are associated with individual licenses which can be found in the metadata.csv file.